Today current papers solve by sir Arslan

(bio301)

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Monosomy

Monosomy is the state of having a single copy of a chromosome pair instead of the usual two copies found in diploid cells

Causes of monosomy: followings are the causes of monosomy

Non disjunction – one gamete receives two copies of homologous chromosomes and other will′ have no copy.

Loss of chromosome. As it move towards pole of cell during anaphase. Fruit fly′

monosomy, is the cause of some human diseases such as Turner syndrome and Cri du Chat syndrome

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What are structural abnormalities?

Structural abnormalities are when part of an individual chromosome is missing, extra, switched to another chromosome, or turned upside down.
Types based on genetic material
Balanced – chromosome complement is complete.′
Unbalanced – when there is incorrect amount of genetic material.′
Types
Translocations′ Deletions′ Insertions′ Inversions′ Ring chromosome′ Isochromosome′

Parents of affected individuals are asymptomatic

In most of the cases, parents of affected individuals are asymptomatic.

deletions

For point **deletions**, one nucleotide has been deleted from the sequence.

cystic fibrosis, Turner syndrome,

Results in frameshift mutation.

Suppressor mutation

Suppresses the effect of other mutation

Occurs at a site different from the site of original mutation.

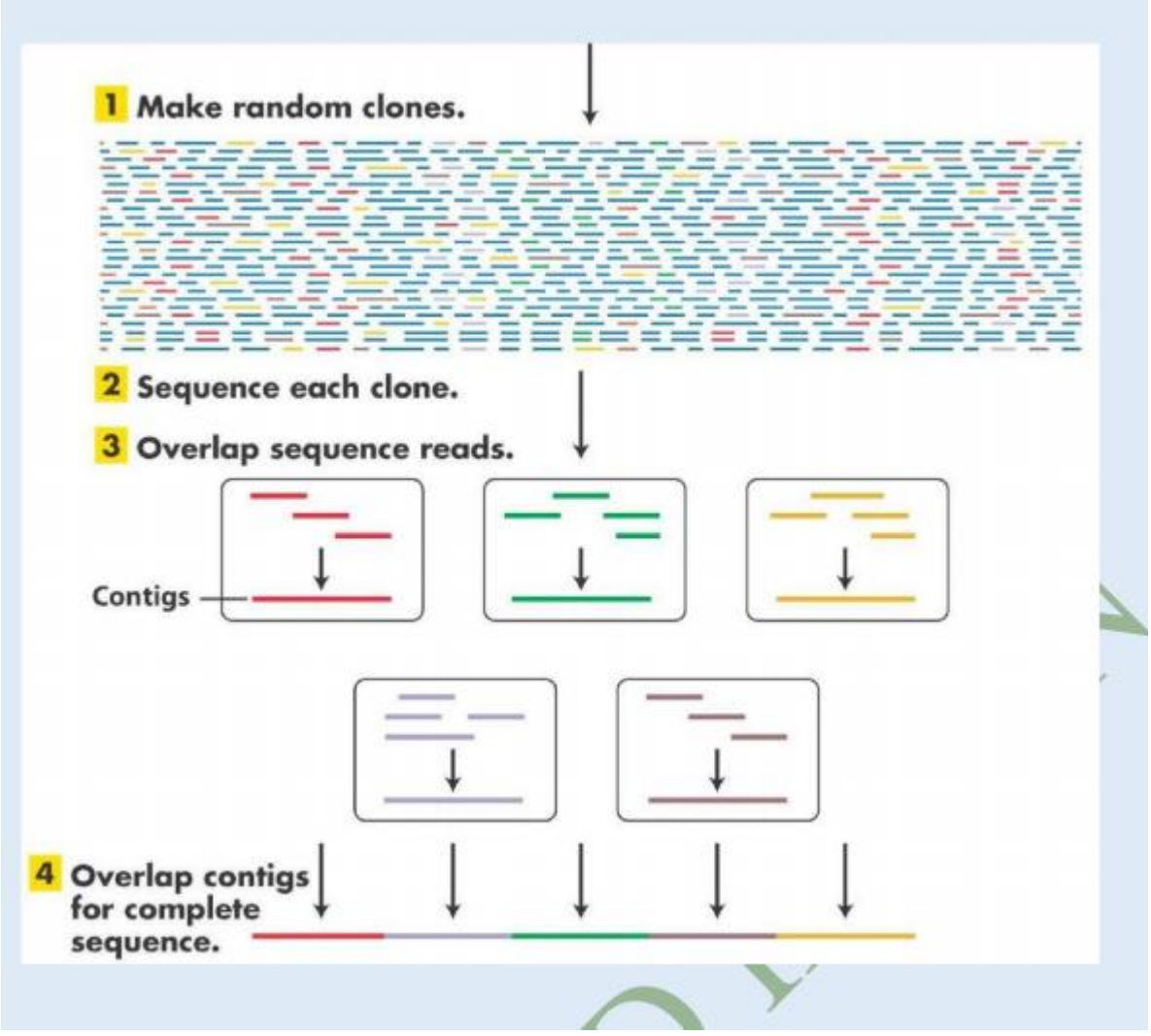
Intragenic suppression

Intragenic suppression results from suppressor mutations that occur in the same gene as the original mutation.

Intergenic suppression

Intergenic (extragenic) suppression relieves the effects of a mutation in one gene by a mutation somewhere else within the genome. Second mutation is not on same gene as original mutation

Bottom-up Mapping



Penetrance and Expressivity

Penetrance measures the proportion of individuals in a population who carry a specific gene and express the related trait. Less than 100% penetrance is a result of modifiers, epistasis, suppressors, and environmental conditions.

In **genetics**, **expressivity** is the degree to which a phenotype is expressed by individuals having a particular genotype. (Alternately, it may **refer to** the expression of particular gene by individuals having a certain phenotype.)

Low and full Penetrant

Low Penetrance An allele with low penetrance will only sometimes produce the symptom or trait with which it has been associated at a detectable level. In low penetrance, it is difficult to distinguish environmental from genetic factors.

Highly Penetrant If an allele is highly penetrant, and then the trait it produces will almost always be apparent in an individual carrying the allele.

Polyploidy

Polyploidy is the heritable condition of possessing more than two complete sets of chromosomes Two major types

Autopolyploidy♠

Allopolyploidy♠

Frequency of autosomal dominant disorders

Frequency of autosomal dominant disorders varies between 3 – 9.5 per 1000 individuals

Frequency of autosomal recessive disorders

Frequency of autosomal recessive disorders is between 2-2.5 per 1000 individuals.

Effect DNA sequence but not protein: Silent mutation. Mcq and definations

Trisomy

Trisomy is a type of polysomy in which there are three copies of a particular chromosome, instead of the normal two. Trisomy is a type of aneuploidy. Trisomy of chromosome 21, which is found in Down syndrome, is called trisomy 21

inversion mutation

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Inversion mutations only affects a small part of the gene.

 May a single nucleotide or more than one nucleotides are involved.

X-linked recessive disorders

Duchene Muscular Dystrophy

Hunter's Disease

Menkes Disease

Hemophilia A and B

Color Blindness

Diagnostic tests for genetic disorders

Physical analysis: Physical analysis for affected individual

Ultrasonography of fetus: Ultrasonography of fetus determines malformations of head, internal organs, and extremities

Analysis of amniotic fluids: Amniotic fluid analysis to determine genetic and chromosomal disorders after 14 wks gestation. More than 200 various genetic diseases easily be detected.

Maternal blood analysis: Maternal blood analysis to observe abnormal fetus.,

Polymorphism

Polymorphism, in biology, a discontinuous genetic variation resulting in the occurrence of several different forms or types of individuals among the members of a single species

Aneuploidy: Change in chromosomes number can occur by the addition/deletion of chromosome or part of a chromosome

. Euploidy: Gain of one or more complete sets of chromosomes

pattern of inheritance

Most genes have one or more versions due to mutations or polymorphisms referred to as alleles. ... There are five basic modes of **inheritance** for single-gene diseases: autosomal dominant, autosomal recessive, X-linked dominant, X-linked recessive, and mitochondrial.

37. Mutation that produce the change in the amino acid in protein product (histidine in for arginine Missense mutation mcq and definition

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Multifactorial diseases six

: Genetic disorders controlled by many genes plus the effects of the environment

Asthma,Cancers ,Diabetes ,Heart diseases,Hypertension,Obesity

Pleiotropy

Pleiotropy The ability of a gene to affect an organism in many ways is called pleiotropy.

 Pleiotropy refers to an allele which has more than one effect on the phenotype.

Types of deletion include the following:

- Terminal deletion – a deletion that occurs towards the end of a chromosome.
- Intercalary/interstitial deletion – a deletion that occurs from the interior of a chromosome.
- **Microdeletion** – a relatively small amount of deletion (up to 5Mb that could include a dozen genes).

Applications of molecular markers – short range

• Parentage determination

• Genetic distance estimation

• Determination of twin zygosity

Applications of Molecular Markers – Long Range

• Gene mapping & mapping of QTL by linkage

• Markers assisted selection of breeds

Method of mapping gene disorders

Genetic maps have been used successfully to find the single gene responsible inherited disorders, like cystic fibrosis and muscular dystrophy.

How genetic mapping performed

Scientists isolate DNA from samples and examine it for the unique patterns of bases.

Molecular techniques are referred to as genotyping polymorphisms or markers.

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